

What Makes Robot Motion Hesitant? Producing and Perceiving Hesitancy in Robotic Manipulators

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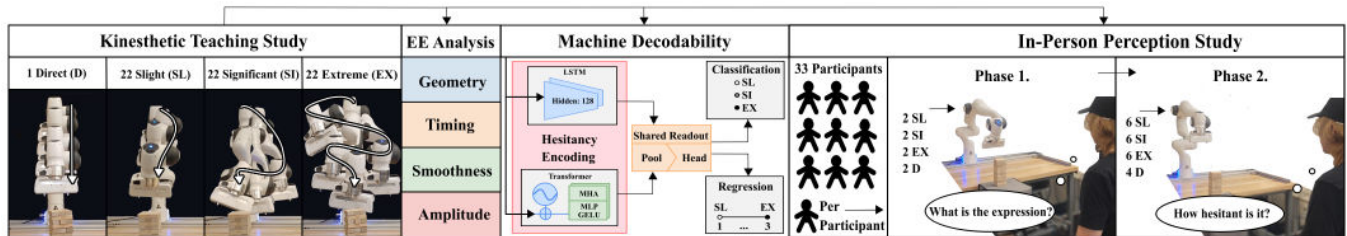


Figure 1: Eleven dancers kinesthetically teach 66 hesitant trajectories at three commanded levels. The recordings feed the end-effector descriptor analysis across four families, the LSTM/Transformer decoding study under leave-dancers-out validation, and a two-phase perception study, where 33 unprimed participants describe motions freely and rate hesitancy levels afterwards.

Abstract

In human-robot collaboration scenarios, *how* a robot moves can be as informative as *what* it is doing. Expressive robot motion has been studied extensively, enabling robots to convey various emotions like happiness, sadness, anger, etc. However, very little attention has been given to task-relevant expressions, where the robot conveys its capabilities and limitations corresponding to its task. When a robot is unsure whether it can succeed at its task, its motion can be shaped to convey hesitancy, which is the behavioral expression of uncertainty in humans. Hesitant robot motion influences how people coordinate with the robot, where they direct attention, and when they intervene. Prior work has pursued this by asking: *Given a fixed robot motion or template, which individual quantities shift perceived hesitancy?* instead of: *When motion is primarily produced to express a commanded level of hesitancy, which quantities change, and how?* This paper answers the latter. In a kinesthetic teaching study, we recruit eleven trained dancers to kinesthetically teach a manipulator to generate 66 free-form targeted trajectories with three commanded levels of hesitancy (i.e., slight, significant, and extreme). Then, we characterize them across four families of end-effector kinematic descriptors (i.e., geometry, timing, smoothness, and amplitude) and observe that a higher hesitancy level is produced as a longer and more circuitous reach with more retreats, while peak speed and acceleration show no detectable change. A model trained on these recordings identifies the commanded hesitancy level ($\kappa = 0.76$ under leave-dancers-out validation), yet a rule that only measures the duration of motion performs just as well. In an in-person perception study, 33 unprimed observers spontaneously attributed hesitancy to these motions and to a slowed

version of a direct trajectory. However, the direct trajectory at its original speed almost never draws that attribution. Once primed, the observers decode the commanded hesitancy level above chance, though less accurately than a duration-only rule. Through these experiments, we observe that the degree of hesitancy is chiefly carried by elapsed time, in production, in machine decoding, and in perception. However, this is mostly due to the dancers traveling further and not due to them slowing down. The form of the motion carries a smaller contribution beyond time, one that dancers produce and observers read but that does not measurably improve machine decoding. Together, these results recast levels of hesitancy from a designed effect into a measured, decodable signal, and point to a starting recipe for principled generation, where elapsed time sets the level, and a retreating, circuitous path is the candidate cue that separates hesitancy from mere slowness.

CCS Concepts

• Human-centered computing → Empirical studies in HCI; Empirical studies in interaction design.

Keywords

Human-Robot Interaction, Expressive Motion, Hesitancy

ACM Reference Format:

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1 Introduction

As robots become widespread in our everyday lives, reliable and effective human-robot collaboration shifts from being a convenience to a necessity. In locations spanning from industrial warehouses to homes, it is imperative that humans feel comfortable and safe around robots, while efficiently achieving a shared goal. Cohesive collaboration requires the robot to clearly communicate what it intends to do and how confident it is in doing it, so that humans can form an accurate mental model of it (i.e., an understanding of its capabilities, goals, and limitations) [33]. Verbal communication

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is the most familiar channel for this, but humans rely on nonverbal cues even when speech is available [7], and equipping robots with such cues strengthens team coherence and cooperation [6, 11]. In robotics, a widely applicable nonverbal cue is motion, which can convey readable signals about a robot’s intentions and confidence. Expressive robot motion has been studied extensively, enabling robots to convey emotions such as happiness, sadness, and anger through movement alone [9, 13, 16, 23, 29, 38]. However, far less attention has been paid to task-relevant expressions, where the robot communicates information about its functionality with respect to the task.

Functional expressivity is defined as the robot’s ability to communicate its functional capabilities and limitations [26]. Among functional states, the expression of *uncertainty* is well understood in humans. When a person is unsure whether an action will succeed, the uncertainty is visible in the action itself, where the motion slows down and the frequency of pauses and retreats increases [20, 24]. This behavioral expression of uncertainty is called *hesitancy*. Robots have no such native expression. Their uncertainty lives in a planner’s confidence estimate that nobody around them can see. This provides us an opportunity to extend the human channel to robots, shaping its motion to convey hesitancy, which is particularly consequential in human-robot collaboration. Whether a robot expresses hesitancy shapes how people coordinate, where they allocate attention, and when they intervene [1, 19, 32]. When a robot is unsure whether it can complete an action, conveying that uncertainty through motion can help nearby humans calibrate trust, anticipate failure, and step in when needed, resulting in a reduction of unsafe surprises and an improvement in teamwork [4, 17, 36]. Like any other expression, hesitancy is not binary. An agent’s level of hesitancy spans a continuous spectrum, where the agent could be slightly unsure, be significantly doubtful, or contemplate avoiding the task completely [12, 25]. Humans routinely identify the individual levels of hesitancy from each other’s movement [7, 10, 24].

For robot motion, prior work has investigated *which quantities of a fixed motion or template shift perceived hesitancy*. It has not asked *which quantities change, and how, when motion is produced to express a commanded level of hesitancy*. This paper answers that question. We present an in-depth analysis of free-form manipulator motion kinesthetically taught by 11 trained dancers, each instructed to express three commanded levels of hesitancy (slight, significant, and extreme). Dancers’ movement expertise provides controlled, repeatable demonstrations while allowing free-form expression [8, 18], giving each trajectory a production-side ground-truth hesitancy level [31]. We then examine which quantities encode the commanded level, how they scale, whether unprimed observers spontaneously attribute hesitancy without being told to look for it, and whether human and machine observers can decode the level without pairwise comparison. Such a characterization recasts the level of hesitancy from a designed effect into a measured, decodable signal, and supplies the empirical foundation that principled generation of hesitant motion requires. In summary, this paper provides two primary contributions:

- Two complementary user studies focusing on the production and perception of 66 free-form hesitant trajectories, kinesthetically taught to a Franka Emika Panda manipulator by 11

trained dancers, where the intensity of the expression is commanded at production (i.e., slight, significant, and extreme) while its form is left free.

- An in-depth analysis of this dataset spanning all three hesitancy levels and all 11 dancers, through end-effector characterization, Transformer and LSTM decoding.

2 Related Work

Robots have expressed hesitancy through multiple modalities, including speech [5, 27, 34], touch [22, 35, 37], and, most relevant to this work, motion [19, 26, 28]. Within motion, one line of work treats hesitancy as a binary expression (i.e., a single hesitant behavior that is present or absent), realized as a hand-designed velocity profile for mobile robots [30], a kinesthetically taught movement quality on a manipulator [26], or a style cost learned from human preferences and imposed through trajectory optimization [39]. A second line pursues the level itself by varying quantities of a fixed robot motion or template to shift perceived hesitancy or conveyed uncertainty [3, 15, 19, 21, 32, 40]. This body of work originates with Moon et al. [20], who recorded hesitant human wrist trajectories and replayed them on a robotic manipulator. Here, the observers recognized the hesitation from the end-effector alone. Furthermore, kinematic analysis of these trajectories produced the Acceleration-based Hesitation Profile (AHP), a template that captured the launch, halt, and retract structure of hesitant motion [21]. However, scaling the template’s launch acceleration across three amplitudes did not change perceived hesitancy. Based on human dyad motion capture [19], the subsequent Negotiative Hesitation Generator (NHG) used kickback distance and re-attempt count to control hesitancy. Likewise, sweeping these parameters failed to yield a monotone intensity scale. In both templates, human hesitation serves as design input rather than as the level-carrying signal itself. Within the same body of work, a robot’s expression of its confidence was controlled through elapsed time. Zhou et al. [40] padded a fixed geometric path with pauses and speed changes to alter how confident a manipulator appeared. Hough and Schlangen coupled a robot’s speed and pre-action wait times directly to its internal confidence estimate, where the observers’ confidence ratings closely followed that estimate [15]. Schömbis et al. adapted this mapping to an in-person collaborative packing task, where participants distinguished between low, high, and full confidence [32]. These results carry two limitations. First, the rated expression is *confidence*, named by the researchers themselves rather than allowing the observers to attribute it spontaneously. In the latter two studies, the robot’s expression was disclosed beforehand [15, 32]. Second, the temporal channel is not decomposed. Speed is either yoked to waiting time [15] or deployed only in combination after proving to be the weakest cue in isolation [32], leaving an open problem of whether speed rather than sheer elapsed time carries the percept. Closest to our work, Bector et al. organized uncertainty-related states in a Commitment-Vigilance space derived from Laban Effort [3], where observers reliably identified which of four states a trajectory expressed. The descriptor values, however, are hand-set inputs, and the four base trajectories are authored by animators to exhibit them, one per state. Furthermore, the intensity of hesitation is assessed pairwise, and reduced approach acceleration reads as

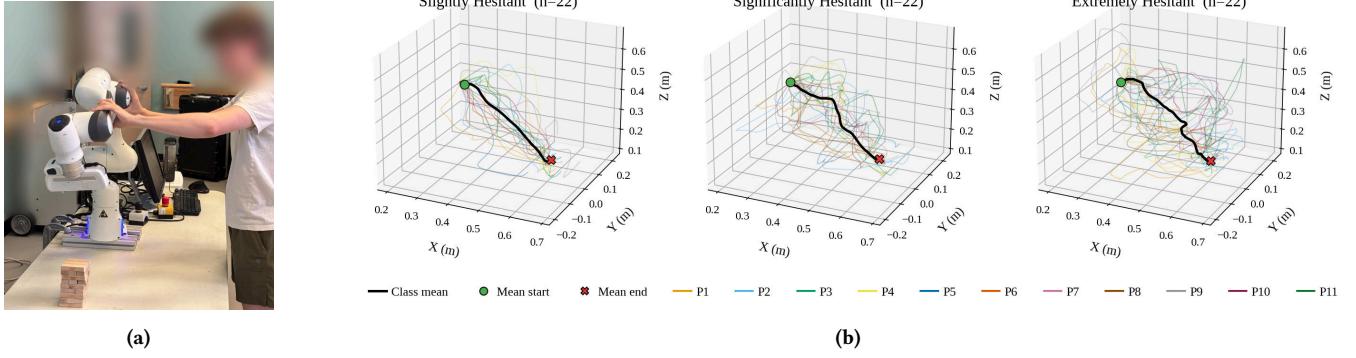


Figure 2: (a) A dancer performing kinesthetic teaching on the Franka Emika Panda manipulator to generate hesitant trajectories towards the Jenga tower at three commanded intensity levels (i.e., slight, significant, and extreme). (b) End-effector trajectories per hesitancy level ($n=22$ each).

both hesitant and fearful. In this paper, the direction of fit is reversed: descriptors are outputs, measured from motion produced to express a commanded level of hesitancy.

Thus, scaling a chosen parameter can test whether that parameter shifts perceived hesitancy, but it cannot identify the carriers of the hesitancy spectrum. Because every other parameter is held fixed by construction, the kinematic signature of increased hesitancy enters these designs as an input rather than emerging as a finding. When the level is commanded, and the trajectory is freely produced to express it, the signature becomes an output, and whether it suffices to recover the level can itself be tested by learned decoders and by unprimed observers. This is what we explore in this paper.

3 Methods

Our methods are organized around three research questions. We first perform a kinesthetic teaching study to collect a dataset of motions produced at commanded hesitancy levels (Section 3.1). Each question is then answered by one component of the study.

RQ1 (Section 3.2): *For a robotic manipulator, which end-effector quantities change when motion is produced at a commanded hesitancy level, and how do they scale across levels?* Prior work characterized human hesitation structurally, as launches, halts, and retractions [20, 21]. We therefore predict that a commanded increase in hesitancy deepens this structural disruption of the reach, rendering it longer, less direct, and less fluent:

- **H1:** Descriptors corresponding to path geometry, timing, and smoothness track the commanded hesitancy level.

Whether the quantities that prior templates scaled without perceptual effect [19, 21] carry the level in this experiment is left to the data rather than predicted in advance.

RQ2 (Section 3.3): *Is the commanded level of hesitancy recoverable from the recorded motion alone?* A descriptor can separate the levels on average while individual trajectories remain ambiguous. Whether a single trajectory is sufficient to identify its level, from end-effector or joint-space signals, is a separate question. We test:

- **H2a:** Learned decoders recover the commanded level above chance for performers unseen during training.

- **H2b:** End-effector representations are at least as decodable as their joint-space counterparts.

RQ3 (Section 3.4): *Do unprimed human observers perceive the commanded hesitancy?* In prior studies, observers either were told beforehand what the motion encoded or compared intensities side by side [3, 15, 32]. Moreover, a decoder can exploit regularities no observer perceives, so machine decodability does not imply human legibility. We test two hypotheses:

- **H3a:** Observers attribute hesitancy spontaneously to the hesitant trajectories rather than to purely time-stretched direct trajectories.
- **H3b:** Observers decode the commanded level above chance, beyond the performance of a duration-only classifier.

Throughout, the end-effector is the unit of analysis, since the dancers shaped it without any embodiment awareness of the joint space. H2b tests this choice from the decoding side.

3.1 Kinesthetic teaching study

We recruited 11 dancers (2 men, 9 women, 9 dancers with 10 or more years of dance training, and 2 dancers with 1–10 years of dance training). The specific genres of the dancers’ expertise spanned Hip-Hop, Ballet, Contemporary, and Jazz. As seen in Figure 2a, we used the Franka Emika Panda robotic manipulator in an IRB-approved kinesthetic teaching setting (gravity-compensation / compliant guidance), where the dancers physically guided the robot from its default configuration towards the Jenga tower. The recruitment was done through emails, where the compensation was \$15 an hour. To isolate expressive differences while keeping the goal constant, the robot always initiated its motion from the same start configuration while the Jenga tower’s pose remained fixed in the workspace. Each dancer performed two takes at each of three commanded levels, *slight*, *significant* and *extreme*, in the same order, resulting in $11 \times 3 \times 2 = 66$ trajectories (Figure 1). For each hesitancy level, the instruction we provided was a single sentence naming the level (“guide the robot to the Jenga tower as if it were expressing [slight / significant / extreme] hesitancy”), with no description of how hesitant motion should look like. The two takes were two attempts at the same instruction, and not rehearsed repetitions. The dancers

were free to vary the form, and the takes are treated as replicates of one commanded level, which is why the dancer-level analysis averages them (Section 3.2). Before instructing the dancers to guide the robot, we showed them a single direct, non-hesitant reference trajectory, kinesthetically guided by the experimenter. Crucially, we told the dancers how hesitant the robot should be, never how to make the robot hesitant. So, the resulting kinematics are a product of the expressive act rather than a parameterization that we imposed. We recorded the `franka_states` ROS topic at 30 Hz, and primarily focused on joint positions q , velocities $\dot{q}$, torques τ and the end-effector pose ${}^O T_{EE}$. Cartesian quantities are evaluated at the end-effector frame (i.e., `panda_grasptarget`), so that the position- and velocity-derived descriptors refer to the same physical point. The end-effector’s linear velocity $\mathbf{v}_{EE} = (J(q) \dot{q})_{1:3}$ is obtained from the robot’s joint velocities through the Jacobian J expressed in world axes, so that $\mathbf{v}_{EE}$ is the world-frame velocity. The acceleration and jerk values of the end-effector are obtained by successive numerical differentiation of $\mathbf{v}_{EE}$ using second-order central differences.

3.2 End-effector cue characterization

After obtaining the dataset, we characterize each trajectory with twelve end-effector descriptors organized into four families: path geometry, timing, smoothness, and amplitude (Table 1). The families encompass multiple facets of targeted motion: where the end effector travels (geometry), how long the motion takes (timing), how fluently it progresses (smoothness), and how intense it is (amplitude). Each trajectory consists of N end-effector position samples $\mathbf{p}_t \in \mathbb{R}^3$, $t = 1, \dots, N$, with Δt taken from the recorded timestamps, and total duration $T = (N-1) \Delta t$. The goal position is $\mathbf{g} = \mathbf{p}_N$, and $\ell_t = \|\mathbf{p}_{t+1} - \mathbf{p}_t\|$ denotes the length of step t . Speed is $v_t = \|\mathbf{v}_{EE}(t)\|$, where $\mathbf{v}_{EE}$, $\mathbf{a}_t$, and $\mathbf{j}_t$ are the end-effector velocity, acceleration, and jerk, while $A = \max_t v_t$ is the peak speed. $\mathcal{A} \subset \{1, \dots, N-1\}$ is the set of steps that move away from the goal by more than $\epsilon = 1 \times 10^{-6}$ m, that is, steps with $\|\mathbf{p}_{t+1} - \mathbf{g}\| > \|\mathbf{p}_t - \mathbf{g}\| + \epsilon$. For spectral descriptors, V_k is the one-sided discrete Fourier transform of the speed profile, with bin index k and frequency resolution Δf . $\hat{P}_k = |V_k|/\max_k |V_k|$ is the magnitude spectrum normalized by its maximum, and f_c is the adaptive cutoff frequency, taken as the frequency of the last bin with $\hat{P}_k \geq 0.05$ within $f \leq 10$ Hz.

Smoothness is estimated in two ways so that the claim does not rest on one formulation. The first is log dimensionless jerk, which grows with every halt, restart, and change of direction. Its T^3/A^2 normalization cancels a uniform rescaling in size or time [14], but not the number of sub-movements. The second is spectral arc length, which lengthens as intermittent corrections add high-frequency content to the speed profile [2]. The acceleration-peak count relies on the norm of the acceleration, so a change of direction registers even when the arm does not slow down. The amplitude descriptors are the quantities earlier template-based work varied directly [21, 32]. We retain them to test whether they carry the hesitancy level when motion is produced freely. Because duration itself rises with hesitancy and mechanically inflates cumulative descriptors, we report every descriptor both raw and with duration partialled out, where the residual from a linear fit on duration is re-tested, and a partial Spearman correlation holds duration constant. Acceleration peaks per second removes duration dependency by construction

Table 1: End-effector descriptors. Color marks the family:
 geometry, timing, smoothness, amplitude.

Descriptor	Definition
Directness ratio	$\ \mathbf{p}_N - \mathbf{p}_1\ / \sum_t \ell_t$
Backtracking fraction	$\sum_{t \in \mathcal{A}} \ell_t / \sum_t \ell_t$
Progress reversals	$ \{t \in \mathcal{A} : t-1 \notin \mathcal{A}\} $
Duration	$T = (N-1) \Delta t$
Log dimensionless jerk [14]	$\ln(\frac{T^3}{A^2} \int_0^T \ \mathbf{j}(t)\ ^2 dt)$
Spectral arc length [2]	$-\sum_k \sqrt{(\Delta f / f_c)^2 + \delta_{k,a}^2}$
Acceleration peaks	$N_{pk} = \text{peaks}(\ \mathbf{a}_t\) _b$
Acceleration peaks/s	N_{pk}/T
Mean speed	$\frac{1}{N} \sum_t v_t$
Peak speed	$\max_t v_t$
Mean acceleration	$\frac{1}{N} \sum_t \ \mathbf{a}_t\ $
Peak acceleration	$\max_t \ \mathbf{a}_t\ $

^a $\delta_k = \hat{P}_{k+1} - \hat{P}_k$, over $f \leq 10$ Hz to the last bin with $\hat{P}_k \geq 0.05$.

^b Peak prominence $\geq 5\%$ of range, min. separation 6 samples, min. width 2.

and asks whether hesitant motion is denser in corrections rather than merely longer.

The dancer, not the trajectory, is the unit of analysis, since a dancer’s two takes per level are not independent. We therefore average each dancer’s takes within each hesitancy level, resulting in 11 values per level. With too few values to verify distributional assumptions, we use rank-based repeated-measures tests, where each comparison is made within a dancer. We use a Friedman omnibus across the three levels, a Wilcoxon signed-rank test of extreme against slight, and a Spearman trend across the ordered levels, with Cohen’s d (population SD across dancers) for the slight-to-extreme contrast and a percentile 95% interval from 4000 bootstrap resamples of dancers. Benjamini–Hochberg correction is applied within each analysis across every descriptor computed in it (nineteen dynamic descriptors, of which Table 1 lists nine, eighteen in the residualized analysis, three path-geometry descriptors). Finally, because a few expressive dancers could drive a population effect, every descriptor is also tested one dancer at a time with an exact sign test on the number of dancers moving in the predicted direction. Here, peak speed, peak acceleration, and mean acceleration, which show no detectable population-level effect, serve as comparison descriptors with a decrease assumed as their predicted direction.

3.3 Machine decodability

Next, we ask whether a learned decoder can recover the commanded hesitancy level, especially from the end-effector alone. We train a two-layer LSTM (hidden 128, dropout 0.3) and a two-layer Transformer encoder ($d_{\text{model}} = 128$, 8 heads), both with mean temporal pooling so the comparison is between architectures rather than readouts, across 22 feature sets (Figure 1). A naive comparison would disadvantage the end-effector, since the recordings carry three joint modalities but only static end-effector pose. Therefore, we derive the end-effector twist and a torque-projected wrench through the arm Jacobian, with linear acceleration and jerk as finite differences of the linear velocity, giving a like-for-like partner to each joint modality. Two comparisons carry the argument: the single-modality velocity channels (end-effector linear and angular velocity against joint velocity) and the 30-dimensional end-effector-only set against

the 21-dimensional joint set. Trajectories are subsampled by two (≈ 15 Hz) to bound the Transformer’s quadratic cost, which hesitancy tolerates as a low-frequency phenomenon. Sequences keep their recorded length, so mean-pooled positional encodings can expose duration to the Transformer for any feature set. Since a dancer’s two takes are near-duplicates, we partition by dancer, so that every test trajectory comes from a performer the model has never seen, and a further subset of *training* dancers is held out for early stopping within each fold. So, the reported test set never participates in model selection. Since folds tile the population, each trajectory is predicted exactly once per seed out-of-fold. We use 5 folds and 5 seeds and report mean and standard deviation across seeds. Training uses Adam (learning rate 10^{-3} , batch 16, weight decay 10^{-4} , gradient clipping 1.0) for up to 100 epochs, stopping early after 20 without validation improvement. Since the level is ordinal, the primary metric is quadratic-weighted κ (chance 0), with accuracy (chance $1/3$). The same ablations are also run as an ordinal regression, checking that the ordering is not an artifact of treating an ordered variable as nominal. Per-dancer decodability re-aggregates the same out-of-fold predictions.

3.4 Perception study

To test whether the kinesthetically taught trajectories are perceived as *hesitant* by human observers, we ran an IRB-approved two-phase study, compensated at \$15 an hour, in which 33 adults watch the physical robot replay recorded trajectories (Figure 1). The participants were recruited through emails and flyers. Hesitant stimuli are drawn in counterbalanced subsets across participants, so that trajectories as well as participants are sampled. The reference trajectory the dancers saw as their anchor serves as the control, presented at its original duration (3.2 s, the *natural* control below) and uniformly time-stretched to approximate the hesitancy class mean durations (7.4, 10.4 and 16.8 s), which is the perceptual counterpart of the duration control in Section 3.2. Each phase begins with unscored practice trials, which are excluded from analysis. The first phase withholds the construct of hesitancy. Participants see eight trajectories (two per level, two direct controls) under a neutral cover story, with the word *hesitancy* never spoken. After observing each trajectory, we ask the participant to describe freely what the robot was expressing, with an explicit “nothing in particular” option. Responses are coded against a four-code codebook (i.e., hesitancy-related, other emotion/attitude/intention, malfunction, nothing) whose categories were fixed before data collection and whose decision rules were fixed before coding began. Two authors performed coding independently from condition-stripped response sheets, with Cohen’s κ for agreement and disagreements resolved by discussion. In the second phase, the construct is revealed via the hesitancy label alone, and we ask the observers to judge the level itself. Here, we execute 22 trajectories (six per level, four direct controls) one at a time and ask the participants to assign a 0–100 hesitancy score plus a categorical level, in randomized class-balanced order. We analyze the first phase with a logistic model fit to the hesitant trials, predicting whether a hesitancy-related attribution occurred from level alone and together with standardized duration. The planned crossed random intercepts for participant and trajectory were estimated at zero. So, we report the logistic fit with 95%

percentile intervals from a joint bootstrap over participants and trajectories (4000 resamples), and per-participant Wilcoxon tests between levels. We analyze the second phase with a linear mixed model on the 0–100 score with crossed random intercepts for participant and trajectory, and a cumulative-link (ordinal logistic) model on the categorical response with intervals from the same bootstrap, both fit to the hesitant trials with level and standardized duration as predictors, alongside per-participant rank correlations and per-level accuracies. An exploratory mixed model pools hesitant and control judgments to compare them at equal duration. Each time-stretched control is compared with its duration-matched hesitant class by a Wilcoxon signed-rank test on per-participant means. Since duration alone is partly diagnostic, human performance is also compared against a duration oracle, a rule that assigns each trajectory the class whose mean duration is nearest, with class means taken from the stimuli themselves. The three duration-matched contrasts are Holm-corrected as one family.

4 Results

4.1 Graded structure in the end-effector path

Across the three commanded hesitancy levels, the mean end-effector path traces a similar course while individual trajectories fan out from a narrow sheaf at *slight* to a dense tangle at *extreme* (Figure 2b). Here, increasing hesitancy does not re-route the trajectory. Rather, it lengthens the exploration around the average path. All three path-geometry descriptors track the commanded level monotonically at the dancer level (Figure 3). The directness ratio falls (trajectory-level median $0.68 \rightarrow 0.49 \rightarrow 0.30$, Friedman $\chi^2(2) = 20.18$, $q < 0.001$, dancer-level Spearman $\rho = -0.70$), the backtracking fraction rises ($0.09 \rightarrow 0.12 \rightarrow 0.28$, $\chi^2(2) = 17.64$, $q < 0.001$, $\rho = +0.54$), and progress reversals multiply ($7.5 \rightarrow 10 \rightarrow 14$; $\chi^2(2) = 14.98$, $q < 0.001$, $\rho = +0.53$). The paired slight-versus-extreme contrast is significant for every descriptor ($p < 0.001$) and every dancer shifts in the predicted direction. With the start and goal fixed, lower directness means a longer path (mean $0.65 \rightarrow 0.86 \rightarrow 1.32$ m, direct reference 0.41 m). Thus, the commanded hesitancy level leaves an ordered imprint on the path itself.

4.2 The cue signature: time and travel

The dynamic descriptors split into a set with large raw effects and a counter-signature set with no detectable effects (Figure 4). The large raw effects are smoothness loss (log dimensionless jerk, trajectory-level means $14.06 \rightarrow 15.51 \rightarrow 17.20$, $d = +3.34$, $q = 0.003$), micro-correction count (acceleration peaks, $10.2 \rightarrow 17.4 \rightarrow 28.6$, $d = +2.84$, $q = 0.003$), and duration ($7.4 \rightarrow 10.4 \rightarrow 16.8$ s, $d = +2.68$, $q = 0.003$). The jerk- and peak-based smoothness measures agree with each other, and spectral arc length trends the same way without surviving correction ($d = -1.06$, $q = 0.067$). Alongside the path-geometry gradients of Section 4.1, every family named in H1 tracks the level, **supporting H1**. The amplitude descriptors are the counter-signature, since peak speed ($d = +0.09$, $q = 0.58$), peak acceleration ($d = -0.19$, $q = 1.00$) and mean acceleration ($d = -0.25$, $q = 0.58$) do not track the level. Mean speed declines only modestly while path length doubles. Thus, a more hesitant reach is longer, less smooth, and more wandering. It is neither faster nor more forceful. Duration control qualifies this claim (Table 2). Partialing

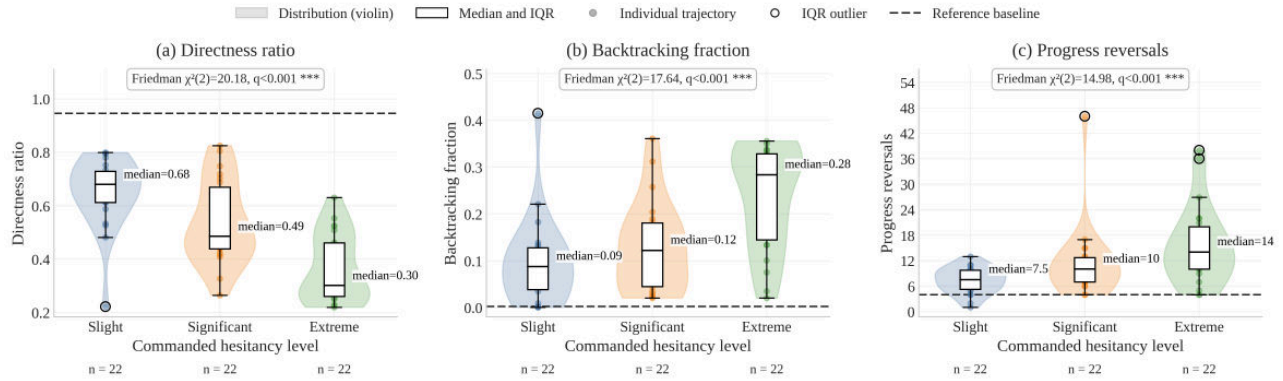
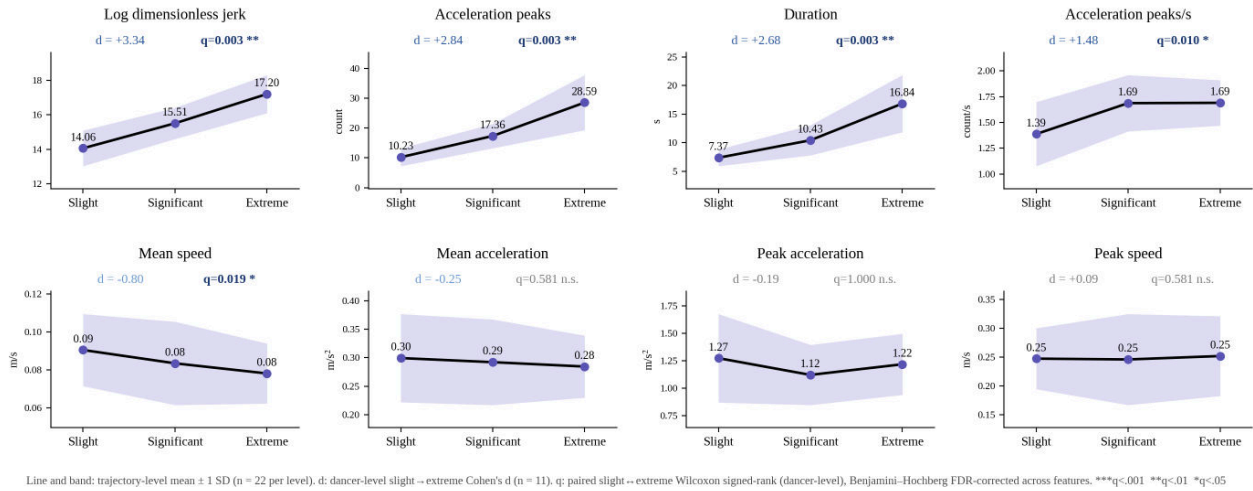


Figure 3: Path-geometry distributions per commanded level. (a) directness ratio, (b) backtracking fraction, (c) progress reversals. Violins show the trajectory-level distribution, while boxes provide the median and the IQR. The dashed line corresponds to the non-hesitant reference trajectory. Annotations report the dancer-level Friedman omnibus (FDR-corrected).



Line and band: trajectory-level mean ± 1 SD ($n = 22$ per level). d : dancer-level slight $\rightarrow$ extreme Cohen's d ($n = 11$). q : paired slight $\rightarrow$ extreme Wilcoxon signed-rank (dancer-level), Benjamini-Hochberg FDR-corrected across features. *** $q<0.001$ ** $q<0.01$ * $q<0.05$

Figure 4: Dynamic descriptors across the three levels, ranked by the dancer-level slight-to-extreme Cohen's d ($n = 11$). Panels display the trajectory-level mean ± 1 SD. d and the FDR-corrected Wilcoxon q are dancer-level. Top row: the descriptors that track the level, including the duration-free rate form. Bottom row: mean speed declines, while the amplitude descriptors show no detectable change.

out duration reduces the jerk and peak-count trends from $+0.83$ and $+0.88$ to $+0.17$ and $+0.48$, and no residualized descriptor survives correction at $n = 11$. Log dimensionless jerk is the clearest case. Across the 66 trajectories it grows as $3.75 \ln T$ ($R^2 = 0.90$), close to the $4 \ln T$ expected when a reach is extended by further sub-movements, so its large raw effect mostly restates duration. Acceleration peaks per second are elevated ($d = +1.48$ [$+0.93, +2.47$], $q = 0.010$) but plateau after the first step ($1.39 \rightarrow 1.69 \rightarrow 1.69$), so correction density separates slight from the two higher levels, but not those from each other. We therefore state the signature as *a longer and more circuitous reach at much the same speed*. Full residualization is a conservative bound, since it also removes signal legitimately expressed through time.

4.3 One vocabulary across teachers

The signature is shared, not idiosyncratic. All three path-geometry descriptors, duration, log dimensionless jerk, and acceleration-peak count move in the predicted direction for all eleven dancers (sign test $p < 0.001$). Acceleration peaks per second and mean speed do so for nine ($p = 0.033$), and spectral arc length for eight ($p = 0.113$). Duration, jerk and peak count are also strictly monotone across the three levels for every dancer, showing that the middle level is not an average of the extremes. The three amplitude controls move in the predicted direction for only four to six dancers ($p = 0.50$ – 0.89).

4.4 Machine decodability

The commanded hesitancy level survives in the part of the motion that an observer can see (Figure 5). **Supporting H2a**, the

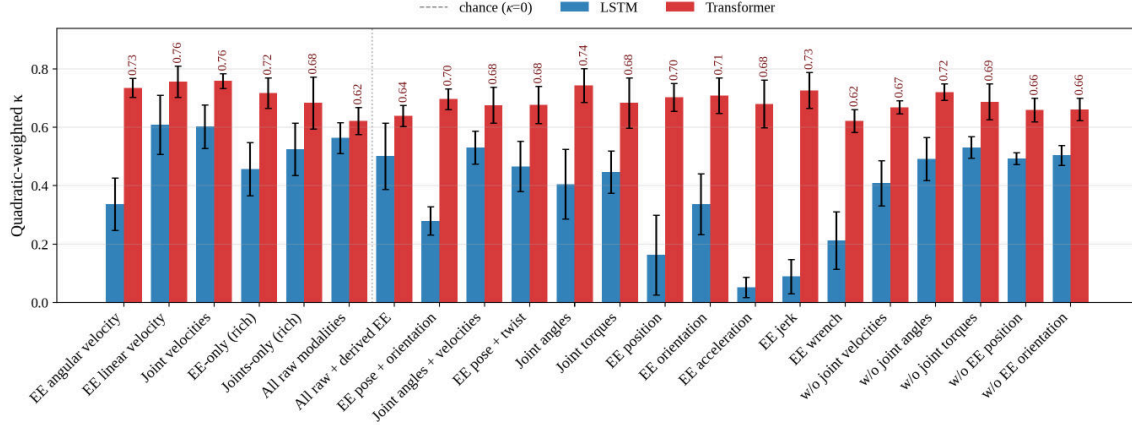


Figure 5: Quadratic-weighted κ (mean $\pm$ SD over five seeds, Transformer values annotated) across all 22 feature sets under leave-dancers-out cross-validation, chance $\kappa = 0$. Left of the divider: end-effector and joint comparisons. Right of the divider: semantic groups, single modalities, and leave-one-out (w/o) variants of the raw set.

Transformer recovers it well above chance from end-effector linear velocity alone ($\kappa = 0.76 \pm 0.05$, accuracy 0.72 against 0.33), and the three velocity channels lie within one seed-SD of one another (0.76, 0.76, 0.73 for joint, end-effector linear and angular velocity), leading us to claim parity rather than an end-effector advantage. The grouped comparison, which matches representational richness, is the cleaner test of H2b. The 30-dimensional end-effector set reaches $\kappa = 0.72 \pm 0.05$ against 0.68 ± 0.09 for the 21-dimensional joint set. The ordinal regression shows the same parity (κ of rounded predictions 0.68 vs. 0.69), and combining all modalities does not help the Transformer. Under the stronger model, the end-effector set is thus no less decodable than the joint set, **partially supporting H2b**, because under the LSTM, the joint set is nominally ahead ($\kappa = 0.46 \pm 0.09$ vs. 0.52 ± 0.09). Separately, per-dancer accuracy for the end-effector Transformer spans 0.37–0.93, so aggregate performance conceals substantial variation across teachers. A duration-only reference bounds these scores, since assigning each trajectory the class with the nearest mean duration, leave-one-dancer-out, already reaches $\kappa = 0.75$ (accuracy 0.73), within one seed-SD of the best decoder. The decoders recover no more level information than elapsed time alone provides. Because mean-pooled positional encodings expose sequence length to the Transformer for every feature set, its runs cannot separate duration from the channels themselves, and the gaps among its feature sets (up to 0.14 in κ , seed SDs 0.02–0.09) do not rank the representations. What the ablation establishes is saturation. No feature set meaningfully exceeds the duration-only rule.

4.5 Human perception

Thirty-three adults participated in the study (age $M = 23.5$, $SD = 5.0$, range 18–39, 16 men, 13 women, 2 non-binary, 1 agender, 1 undisclosed) with each session lasting 40–45 minutes. Self-rated robot familiarity was moderate (median 3 of 5, 9 of the 32 who answered rated it 4 or above), all reported normal or corrected-to-normal vision, and none had taken part in the authors’ previous studies. One second-phase trial (participant 1, trial 15) is excluded

Table 2: Does the cue survive duration control? Spearman trend before (ρ) and after ($\rho_{|T}$) partialing out duration, and the slight-to-extreme Cohen’s d before (d) and after (d_{res}) residualizing duration, with the residualized FDR-corrected Wilcoxon q_{res} , all at the dancer level ($n = 11$). No residualized effect survives correction ($q_{res} > 0.45$).

Descriptor	ρ	$\rho_{ T}$	d	d_{res}	q_{res}
Log dimensionless jerk	+0.83	+0.17	+3.34	+0.77	0.62
Acceleration peaks	+0.88	+0.48	+2.84	+0.66	0.46
Acceleration peaks/s	+0.48	+0.39	+1.48	+0.81	0.46
Spectral arc length	−0.37	+0.04	−1.06	+0.63	0.46
Mean speed	−0.28	−0.01	−0.80	−0.16	0.72

because of a corrupted stimulus file, repaired before all later sessions. In total, 725 of 726 second-phase judgments (593 of 594 on hesitant trajectories) and all 264 first-phase responses remain.

Phase 1. Here, hesitancy is attributed to the natural control once in 33 trials (Figure 6a), while it rises across the hesitant levels ($0.30 \rightarrow 0.33 \rightarrow 0.64$), and is also drawn by the 10 s stretched control (0.36). The remaining responses shift from “nothing” toward other attributed states as the level rises (nothing: 0.39 for the natural control, $0.17 \rightarrow 0.09 \rightarrow 0.05$ across the hesitant levels), and malfunction is almost never read into the motion (3 of 264 responses). Merely slowing a motion invites hesitancy readings. Among hesitant trials, attribution rises with level (level entered alone, odds ratio 2.0 [1.1, 4.3] per step), driven by the extreme level (per-participant Wilcoxon, extreme vs. slight and vs. significant both $p = 0.003$, significant vs. slight $p = 0.70$). Level and duration are too collinear in these 198 binary responses ($r = 0.75$) to be separated (entered together, level 0.37 [−0.66, 1.48] log-odds per step, duration 0.38 [−0.46, 1.36] per SD). The rise from slight to extreme survives scoring every coder disagreement against the hypothesis ($0.21 \rightarrow 0.26 \rightarrow 0.50$, Wilcoxon $p = 0.005$) or for it ($0.38 \rightarrow 0.50 \rightarrow 0.76$, $p < 0.001$), and excluding the one participant whose responses a coder had previously seen ($\kappa = 0.58$, rates within 0.02).

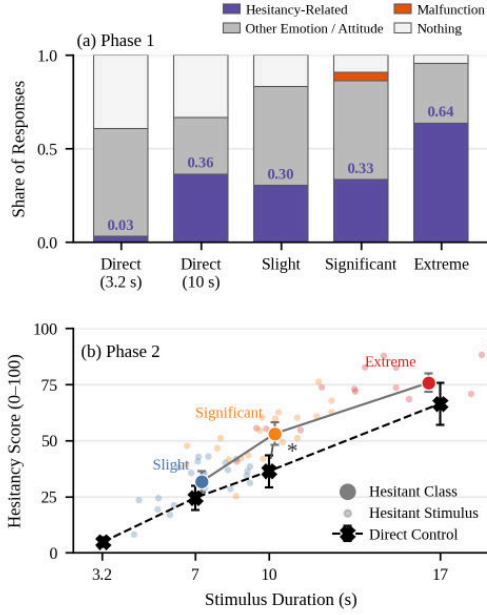


Figure 6: Perception study ($n = 33$). (a) Phase-1 response composition per condition (resolved codes, all 264 responses, coder agreement $\kappa = 0.57$), with the hesitancy-related share annotated. (b) Phase-2 hesitancy score against stimulus duration: condition means with 95% CIs over participants, direct controls (crosses, dashed) and hesitant classes (filled, solid) plus the 66 per-stimulus means (faint dots), the asterisk marks the one duration-matched contrast that is significant (Holm $p = 0.002$).

Phase 2. Participants ordered the commanded levels (Figure 6b). Here, scores rise $32 \rightarrow 53 \rightarrow 76$, the median within-participant Spearman correlation between score and level is $\rho = +0.73$ (Wilcoxon vs. zero, $p < 10^{-9}$), and categorical decoding reaches accuracy 0.52 against 0.33 chance (quadratic-weighted $\kappa = 0.56$, 7% of judgments miss by more than one level). Elapsed time is the dominant carrier. The natural direct control scores 5, while its 7, 10 and 17 s stretches score 25, 36 and 66, and a duration oracle beats pooled human accuracy (0.76 vs. 0.52, 31 of 33 participants fall below it, $p < 10^{-6}$). However, duration is not the whole story. With duration held fixed, level still adds +13.7 [8.8, 18.5] and +20.1 [13.1, 27.0] points for significant and extreme relative to slight (linear mixed model, duration +13.1 [10.3, 16.0] per SD, all $p < 0.001$). The estimates shrink but hold when duration enters on a log scale, which fits the scores better (+9.3 [4.4, 14.1] and +15.4 [8.4, 22.4], both $p < 0.001$), and the cumulative-link model agrees (level 0.83 [0.31, 1.39] log-odds per step, duration 1.16 [0.69, 1.85] per SD). At matched durations, the significant class out-scores its 10 s control (53 vs. 36, Holm $p = 0.002$), while the slight (32 vs. 25) and extreme (76 vs. 66) contrasts run the same way without reaching significance ($p_{\text{Holm}} = 0.16$ each).

H3a is **supported against the natural control and not against the stretched one**. Attribution is spontaneous and rises with level,

but slowed direct motion draws attributions as often as the slight and significant classes, and only the longer extreme class exceeds it (0.64 vs. 0.36, post hoc $p = 0.003$). H3b is **supported in its first clause and not its second**. Observers decode the level well above chance but fall short of a duration oracle. Phase 1 cannot separate level from duration, while Phase 2 can. Among hesitant trajectories, level adds to the score beyond duration, and hesitant motion out-scores direct motion slowed to the same duration (by +7.9 to +12.8 points, every interval above zero). Thus, perceived hesitancy is primarily a temporal code, with form contributing to the degree. What carries that contribution remains open. Across the 66 stimulus means (593 judgments, 11 dancers), neither correction density nor path length predicts the score once level and duration are accounted for ($p > 0.6$), and only backtracking fraction does under both duration scalings (+2.9 to +3.6 points per SD, $p = 0.01$ –0.03, uncorrected, exploratory).

5 Conclusion and Future Work

In this work, we investigated how a commanded level of hesitancy is expressed in the kinematics of a manipulator’s motion, and whether unprimed observers recognize it. We observed that the level of hesitancy lives in the shape and timing of the reach: each step up makes it longer in time and in distance, less direct, and less fluent. Most of the added time is due to added travel rather than slowing or pausing. Peak speed and acceleration, the quantities earlier templates scaled, show no detectable change. All eleven dancers share this vocabulary, and a single trajectory from an unseen performer reveals its level, though no better than its duration alone. Unprimed observers attribute hesitancy to these motions spontaneously and rarely to the direct reference trajectory at its original speed. Once the construct is named, they decode the level well above chance, though less accurately than a rule based on elapsed time alone. Their ratings still carry level information beyond duration, and they provide a higher rating for hesitant motion over direct motion that is slowed down to the same duration.

These findings describe one manipulator, one reaching task, and eleven movement experts whose legibility varies. The three levels were performed in a fixed order, so each may have been produced relative to the one before it, which could exaggerate their separation. Durations showed no reliable drift between first and second takes (+0.3 s, $p = 0.25$), but only a counterbalanced replication can rule out anchoring. The near-constant peak speed may partly reflect the comfortable pace of hand-guiding a compliant arm. Only the attribution phase of the perception study is unprimed, merely slowed motion already invites attributions, and the uniformly stretched controls confound mean speed with elapsed time.

For the field, this work influences how hesitant robot motion is generated. Designers gain a starting recipe: elapsed time sets the degree, and the dancers fill that time mostly with a circuitous, retreating path rather than with reduced speed or pauses. Rather than scaling a hand-picked parameter, generative models can be conditioned on the signature that people produce and read, and evaluated with the same descriptors and decoders. Finally, human–robot interaction gains a framework that transfers to other functional states that a robot can communicate through motion.

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